Automaton

A new user needs a new OS.

The problem

We've gotten to the point—in the AI era—where the models are not the problem. Two years ago, the best model completed 12% of everyday tasks on a real desktop (the OSWorld benchmark). The best now completes 86%, more than the 72% a person does. Open-source models are only about four months behind the frontier, and they are more than good enough for most tasks carried out day to day. Yet the way the world works has hardly changed: only about 10% of American firms use AI to produce what they sell. Companies and VCs have concluded that we have an application problem, and they are rushing to find and build applications. However, every application runs on an operating system, and that is where the obstacle is.

The conceptual and practical design of the operating system was made in an age where humans and humans alone were its users: a person at a screen, with a keyboard and a mouse. Today, we've invented a technology—AI—that understands these systems better than we do, but is bottlenecked by them far more than we are. The system has no place for an agent, so the agent borrows a person's screen, account, and files.

I collected 1,214 public reports of agents failing on today's systems and sorted 1,168 of them by cause. Almost none blame the model. They blame the system around it: isolation bolted on from outside, agents holding their user's full authority, processes nobody owns or can stop, agents colliding on one desktop, and work destroyed with no undo. In my own test, a real agent read a planted secrets file and repeated it 17.4 seconds after it started. Nothing stopped it. The full study is published at kdronald3.github.io/aios-agent-os-pain.

Companies have realized this and have decided to build applications on the cloud, where they can redesign the foundations. However, there is only so much you can do on the cloud, given the main user base lives in the real world, and their data follows them there. Organizations, laboratories, people, and the government all have data they'd rather not share. We need to rework the operating system with the recognition that there is a new user around that is nothing like the original one.

What exists today

Automaton is an open-source operating system, based on Linux, that treats an agent as a second kind of user with a place of its own, instead of a program that runs under your name. I built a working demo in three days and 200 commits, on an 8 GB machine with no graphics card, before any funding, working with AI agents.

On today's systems, only one agent can use the computer at a time, and you cannot use it while it does. On Automaton:

- Six agents in six ordinary apps finished six long tasks in one run of 22 minutes, each on its own desktop.
- I kept typing beside them: a key press reached the screen in 31 to 43 ms, against 24 to 26 ms with no agents.
- One more agent costs 221 MB of memory.

A three-minute recording of that run: youtu.be/Ugkh9cJ_mVg

How it works

Today an agent is an app that runs as you. On Automaton it is a user of the system.

Working today:

- **Screen.** Each agent has its own desktop. You keep yours.
- **Files.** An agent writes to its own layer, and you keep or discard what it did.
- **Watching.** Every agent's desktop is on one live view.

Being built in stage one:

- **Identity.** An agent has its own account, with only the access you grant.
- **Secrets.** They are used through the system and never read by the agent.
- **When you leave.** The agent works on it, and you reach it from your phone.
- **Models.** Local or cloud, run by the system.

Of course, one question stands. If this is a new OS, how do users keep their existing workflows? Automaton runs Windows applications on its Linux base, an approach Valve has already proven. I tested 697 Windows applications drawn at random: 46% already run, and 66% run at least in part. Every failure is traced to its cause. This is where cheap execution helps: with a lean team, AI agents will be employed heavily to close the gap. This part is crucial and cannot be compromised on.

The next 18 months

Stage one is the redesign of the OS for laptops and desktops, on the machines people already own. There are no users yet, and I am the only founder; I am looking for one or two co-founders, then engineers, for a team of five. The goal is to take Automaton from a working demo to a system its first pilot customers use every day.

- **Month 3:** The first co-founder or engineer has joined. The demo is hardened.
- **Month 6:** Version one of the seat layer on Linux. Security review started.

- **Month 9:** The team is five. Local models drive agents, measured.
- **Month 12:** Automaton installs as its own system. The first pilot pays.
- **Month 18:** The pilots' Windows applications run.

I am ready to change the plans to suit the needs that I find as I work on this.

Why now?

Why can a new OS succeed today? Unlike previous attempts at building new operating systems, the new user—AI—makes it possible to give people an experience that cannot be gotten from the alternatives. And the established companies have three major problems.

- **Their revenue depends on today's design.** Apple's services earned $123 billion last year, a quarter of the company, much of it from apps sold to people. A system where agents do the work cuts into that. Apple and Google can't afford to lose that revenue while they wait a few years for a new stream.
- **They cannot break what a billion people use.** Windows runs on 1.6 billion devices, and every change has to leave all of them working. So anything new arrives slowly and optional: Microsoft announced Recall, withdrew it within four weeks of the reaction, and shipped it eleven months later, switched off by default.
- **Their users do not trust them with it.** When Microsoft said Windows was becoming an "agentic OS," it drew 430 replies, nearly all hostile, and later dropped the phrase. This is why Automaton has to be open source: anyone can read what it does.

Microsoft's own attempt, Agent Workspace, has been a preview since 2025: switched off by default, on Windows only, with no view of several agents at once and no undo of what an agent changed. A new company, however, is like a seedling: it is in a unique spot because it is still growing. It will grow through the cracks to chase the sunlight until it is tall enough to compete with the established trees around it, and by then it will be too late for the incumbents because of the trust and reputation the new company will have garnered along the way.

Where the revenue comes from

The OS will be open source. It has to be, to gain any trust, and it is how the OS gets onto people's laptops. In the early days the revenue will come from companies: an enterprise edition, at $30 per person per month with agents included, that lets them use local and cloud models at full capacity without compromising their data.

For a 500-person firm running three agents each, cloud PCs for agents cost $438,000 a month if they stay on, or $105,600 for working hours only (at Microsoft's price of $0.40 an hour for Windows 365 for Agents). Automaton, on PCs the firm already owns, costs $15,000.

The long-term vision

AI has made execution cheap, so a small product can be copied in a weekend. An operating system cannot. It takes compatibility with the applications people already use, a security record, and the trust of the people who rely on it. That is why the plan is this large.

- **Stage two: mobile devices.** Phones, watches, and headsets are the devices people actually use in their daily lives, and bringing the OS close to them lets the application layer benefit.
- **The stores.** Perhaps in the future people won't be buying apps, but tools that give their agents access to different data, and models tested for safety that they can buy or rent for bespoke use cases. The OS has to be the distribution center for all of these.
- **Hardware.** With data on how agents work and how people use their computers, we can design hardware best suited to the new user base.
- **Our own models.** Only if the data says so.